

Arjuna JEE (2025)

Maths

DPP: 3

Trigonometric Identities

Q1 The value of $\frac{1-\tan^2 15^\circ}{1+\tan^2 15^\circ}$ is:

- (A) 1
(B) $\sqrt{3}$
(C) $\frac{\sqrt{3}}{2}$
(D) 2

Q2 Proof :

$$\tan A + \cot A = 2 \operatorname{cosec} 2A.$$

Q3 Proof that:

$$\tan A - \cot A = -2 \cot 2A$$

Q4 Proof that:

$$\operatorname{cosec} 2A + \cot 2A = \cot A$$

Q5 Proof that:

$$\frac{1 + \tan^2(45^\circ - A)}{1 - \tan^2(45^\circ - A)} = \operatorname{cosec} 2A.$$

Q6 If $0 < \theta_2 < \theta_1 < \frac{\pi}{4}$, $\cos(\theta_1 + \theta_2) = \frac{3}{5}$ and $\cos(\theta_1 - \theta_2) = \frac{4}{5}$, then $\sin 2\theta_1$ equal to

- (A) -1 (B) 1
(C) 2 (D) -2

Q7 $\frac{\tan(\frac{\pi}{4} + x)}{\tan(\frac{\pi}{4} - x)}$ is equal to

- (A) $\left(\frac{1+\tan x}{1-\tan x}\right)^2$
(B) $\left(\frac{1-\tan x}{1+\tan x}\right)^2$
(C) $\left(\frac{1+\sin x}{1-\sin x}\right)^2$
(D) $\left(\frac{1-\sin x}{1+\sin x}\right)^2$

Q8 If $\cos(\alpha + \beta) = \frac{4}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and α, β lie between 0 and $\frac{\pi}{4}$ then find $\tan 2\alpha$

- (A) $\frac{56}{33}$
(B) $\frac{33}{56}$
(C) $-\frac{56}{33}$
(D) $-\frac{33}{56}$

Q9 If $\tan \frac{\theta}{2} = t$, then $\frac{1-t^2}{1+t^2}$ is equal to

- (A) $\cos \theta$
(B) $\sin \theta$
(C) $\sec \theta$
(D) $\cos 2\theta$

Q10 If $90^\circ < A < 180^\circ$ and $\sin A = \frac{4}{5}$, then $\tan \frac{A}{2}$ is equal to

- (A) $\frac{1}{2}$
(B) $\frac{3}{5}$
(C) $\frac{3}{2}$
(D) 2

Q11 If $\sin \alpha = \frac{-3}{5}$ where $\pi < \alpha < \frac{3\pi}{2}$, then $\cos \frac{\alpha}{2}$ equal to

- (A) $\frac{1}{\sqrt{10}}$
(B) $-\frac{1}{\sqrt{10}}$
(C) $\frac{3}{\sqrt{10}}$
(D) $-\frac{3}{\sqrt{10}}$

Q12 If $\pi < \theta < \frac{3\pi}{2}$ and $\cos \theta = -\frac{3}{5}$, then $\tan(\frac{\theta}{4})$ is equal to

- (A) $\frac{\sqrt{5}-1}{2}$
(B) $\frac{\sqrt{5}+1}{2}$
(C) $\frac{-\sqrt{5}+1}{4}$
(D) $\frac{-\sqrt{5}-1}{4}$

Q13 If $\sin 2\theta = \frac{3}{4}$, then $\sin^3 \theta + \cos^3 \theta =$

- (A) $\frac{\sqrt{5}}{8}$
(B) $\frac{\sqrt{7}}{8}$
(C) $\frac{\sqrt{11}}{8}$



(D) none of these

Q14 The value of $\cos^3\left(\frac{\pi}{8}\right)\cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)$ is

- (A) $\frac{1}{\sqrt{2}}$
 (B) $\frac{1}{2}$
 (C) $\frac{1}{4}$
 (D) $\frac{1}{2\sqrt{2}}$

Q15 If $\cos(\alpha + \beta) = 3/5$, $\sin(\alpha - \beta) = 5/13$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to

- (A) $21/16$ (B) $63/52$
 (C) $33/52$ (D) $63/16$

Q16 The value of $(1 + \cos \frac{\pi}{8})(1 + \cos \frac{3\pi}{8})(1 + \cos \frac{5\pi}{8})(1 + \cos \frac{7\pi}{8})$ is:

- (A) $\frac{1}{4}$
 (B) $\frac{3}{4}$
 (C) $\frac{1}{8}$
 (D) $\frac{3}{8}$

Q17 $8 \sin \frac{x}{8} \cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8}$ is equal to

- (A) $\sin x$ (B) $8 \sin x$
 (C) $\cos x$ (D) $8 \cos x$



Answer Key

Q1 (C)

Q2 Hence Proved

Q3 Hence Proved

Q4 Hence Proved

Q5 Hence Proved

Q6 (B)

Q7 (A)

Q8 (A)

Q9 (A)

Q10 (D)

Q11 (B)

Q12 (B)

Q13 (D)

Q14 (D)

Q15 (D)

Q16 (C)

Q17 (A)



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